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CorNet: Autonomous feature learning in raw Corvis ST data for keratoconus diagnosis via residual CNN approach

PeiPei Zhang [a,1], LanTing Yang [a,1], YiCheng Mao [a], XinYu Zhang [a], JiaXuan Cheng [a], YuanYuan Miao [a], FangJun Bao [a,c], ShiHao Chen [a,c,**], QinXiang Zheng [a,c,***], JunJie Wang [a,b,d,*]

[a] School of Ophthalmology and Optometry, Eye Hospital, Wenzhou Medical University, Wenzhou, 325027, China
[b] National Engineering Research Center of Ophthalmology and Optometry, Eye Hospital, Wenzhou Medical University, Wenzhou, 325027, China
[c] National Clinical Research Center for Ocular Diseases, Eye Hospital, Wenzhou Medical University, Wenzhou, 325027, China
[d] Department of Ophthalmology, The Third Hospital of Mianyang, Sichuan Mental Health Center, Mianyang, 621054, China

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ABSTRACT

Purpose: To ascertain whether the integration of raw Corvis ST data with an end-to-end CNN can enhance the diagnosis of keratoconus (KC).
Method: The Corvis ST is a non-contact device for in vivo measurement of corneal biomechanics. The CorNet was trained and validated on a dataset consisting of 1786 Corvis ST raw data from 1112 normal eyes and 674 KC eyes. Each raw data consists of the anterior and posterior corneal surface elevation during air-puff induced dynamic deformation. The architecture of CorNet utilizes four ResNet-inspired convolutional structures that employ 1×1 convolution in identity mapping. Gradient-weighted Class Activation Mapping (Grad-CAM) was adopted to visualize the attention allocation to diagnostic areas. Discriminative performance was assessed using metrics including the AUC of ROC curve, sensitivity, specificity, precision, accuracy, and F1 score.
Results: CorNet demonstrated outstanding performance in distinguishing KC from normal eyes, achieving an AUC of 0.971 (sensitivity: 92.49%, specificity: 91.54%) in the validation set, outperforming the best existing Corvis ST parameters, namely the Corvis Biomechanical Index (CBI) with an AUC of 0.947, and its updated version for Chinese populations (cCBI) with an AUC of 0.963. Though the ROC curve analysis showed no significant difference between CorNet and cCBI ($p = 0.295$), it indicated a notable difference between CorNet and CBI ($p = 0.011$). The Grad-CAM visualizations highlighted the significance of corneal deformation data during the loading phase rather than the unloading phase for KC diagnosis.
Conclusion: This study proposed an end-to-end CNN approach utilizing raw biomechanical data by Corvis ST for KC detection, showing effectiveness comparable to or surpassing existing parameters provided by Corvis ST. The CorNet, autonomously learning comprehensive temporal and spatial features, demonstrated a promising performance for advancing KC diagnosis in ophthalmology.

1. Introduction

Keratoconus (KC) is a bilateral, progressive, and asymmetric ectatic corneal disease characterized by localized biomechanical deterioration, stromal thinning, and subsequent corneal protrusion, which causes irregular astigmatism and visual impairments [1,2]. At the heat of corneal refractive surgery, undetected KC are believed to be highly associated with iatrogenic keratectasia, the most severe and irreversible complication after the surgery [3]. Therefore, the accurate identification of KC is necessary during preoperative screening for refractive surgery. Accurate and timely detection of KC is also vital in delaying or stopping its progression using therapies such as corneal cross-linking that

* Corresponding author. No. 270 XueYuan West Road, WenZhou City, ZheJiang Prov, 325027, China.
** Corresponding author. No. 270 XueYuan West Road, WenZhou City, ZheJiang Prov, 325027, China.
*** Corresponding author. No. 270 XueYuan West Road, WenZhou City, ZheJiang Prov, 325027, China.
E-mail addresses: csh@eye.ac.cn (S. Chen), qinxiangzheng@wmu.edu.cn (Q. Zheng), junjie.wang@wmu.edu.cn (J. Wang).
[1] PeiPei Zhang and LanTing Yang contributed equally to this work.

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strengthens the cornea [4].

Current clinical assessment and diagnosis of KC mainly rely on corneal morphology (including topography and tomography) and biomechanical evaluations. Corneal tomography is the most commonly used tool to access corneal morphology and is crucial for KC diagnosis, yet it may yield false negatives in the early stages of KC [5]. Meanwhile, recent evidence suggests that corneal biomechanical changes precede visible morphological alterations, emphasizing the importance of biomechanical analysis for early KC detection [6,7]. Analysis of corneal biomechanical properties brings with it the prospects of revealing the cause (biomechanical weakening) possibly much earlier than the effect (morphological changes), therefore allowing a timelier diagnosis [8,9]. However, in vivo measurement of corneal biomechanics is still facing some unsolved challenges.

The Corvis ST is a one of the two clinical devices currently available for measuring corneal biomechanics in vivo (the other is Ocular Response Analyzer, ORA). It utilizes an ultra-high-speed Scheimpflug camera to record the dynamic corneal deformation after a rapid air-puff excitation to the front cornea, providing richer information on the corneal biomechanics than the ORA. The analysis of this deformation generates a series of dynamic corneal response (DCR) parameters to represent the in-vivo corneal biomechanical properties. While showing promising diagnostic potentials in several studies [10–12], direct KC detection through DCR parameters still lacks standardized criteria and defined cut-off values. Recent developments of Corvis ST have introduced several combined indices such as the Corvis Biomechanical Index (CBI) [13], the Corvis Biomechanical Index for Chinese populations (cCBI) [14], and the Tomographic Biomechanical Index (TBI) [15] which use machine learning (ML) algorithms to integrate various DCR parameters (CBI and cCBI) and with tomography data (TBI). Although these additional indices were reported to improve KC detection, there may be cases where predictions of CBI and TBI differ [16].

The possible drawbacks underlining current Corvis ST paramaters may lies in their reliance on low-dimensional data in the form of DCR parameters. These parameters, engineered to capture specific features, are usually extracted from pre-defined regions of the cornea and a handful of time-stages during its dynamic deformation [17], thus might overlook vital information embedded in the raw measurement data for accurate KC detection [18,19]. As a biological soft tissue, the corneal material behaviours are viscous and hyperelastic (viscoelasticity) and regionally variable [20,21]; however, current DCR parameters may be limited in capturing comprehensive spatial and temporal information of the deforming cornea that reflect its regionally varing viscoelasticity. Efficiently harnessing these spatial and temporal information may therefore identify local changes in corneal viscoelasticity and thus further improve KC detection.

Various diagnostic algorithms, such as decision trees, discriminant analysis, and neural networks, have combined Corvis ST and ML techniques [22–24]. Although these traditional ML algorithms have exhibited promising diagnostic performance in certain datasets, they still encounter limitations due to their reliance on low-dimensional data, as previously mentioned. Convolutional Neural Network (CNN), an advanced ML technique, has been gaining traction in ophthalmology in recent years. Unlike traditional ML methods relying on specifically engineered and a priori selected features, CNN accepts raw or minimally preprocessed data as inputs, create features automatically without bias from theoretical assumptions [25]. This is crucial when dealing with the substantial raw data from Corvis ST and can be beneficial for taking in both spatial and temporal information of the dynamic corneal deformation. To the best knowledge of the authors, there is limited prior research utilizing a CNN approach to analyze raw Corvis ST data for detecting KC [26].

In this paper, a novel approach, called CorNet, was introduced to diagnosing KC with CNN which utilizes the full potential of raw Corvis ST data. The CorNet comprises four convolutional blocks with a residual structure similar to Residual Neural Network (ResNet) [27], integrating 1×1 convolution in identity mapping to ensure a lightweight network. To increase the interpretability of the model, Gradient-weighted Class Activation Mapping (Grad-CAM) [28] was employed to visualize the average attention allocated to diagnostic regions. The contributions of this work can be summarized as follows.

- First, this study proposes an end-to-end CNN utilizing the raw data obtained by the Corvis ST for KC detection, instead of the DCR parameters as in the previous methods.
- Second, CorNet, a novel structure based on the basic residual block, is capable of autonomous feature learning to capture comprehensive temporal and spatial features from raw data.
- Finally, the model evaluations indicated that CorNet can efficiently detect KC, showing diagnostic efficacy comparable to or better than the currently available Corvis ST parameters.

The remaining part of the study are structured as follows: Section 2 discusses the methods employed in the study, including the experimental setup and data analysis procedures. Section 3 presents the results of the experiments, followed by a discussion of the findings in Section 4. Finally, Section 5 presents the overall conclusions of the study.

2. Method

2.1. Participants

This retrospective study complied with the principles of the Declaration of Helsinki and was approved by the Institutional Review Board of Wenzhou Medical University. Written consent form was obtained from each subject. Complete ocular examinations were performed, including visual acuity, objective and manifest refraction, slit-lamp microscopy, noncontact intraocular pressure, fundoscopy, corneal morphology, and corneal biomechanics. Patients were classified into the KC or normal groups as follows:

- KC was defined by the presence of one of the following clinical signs: stromal thinning, Fleischer's ring, Vogt's striae, Munson's sign, epithelial or subepithelial scarring associated with a typical topographic pattern (asymmetric bow tie with a skewed radial axis), central K-value $>$ 47.00 diopters (D) and inferior-superior asymmetry (I–S value) $>$ 3.0 D according to Rabinowitz's criteria and a corrected distance visual acuity $<$ 1.0. All participants had no history of wearing contact lenses.
- Normal eyes were randomly selected candidates for corneal refractive surgery with no history of ocular surgery or trauma, no active ocular disease, and a corrected distance visual acuity $\geq$ 1.0. Contact lenses were discontinued (soft contact lenses for at least 2 weeks, rigid contact lenses for at least 4 weeks, and keratoplasty lenses for at least 3 months) before corneal assessments.

2.2. Corvis ST data acquisition

Corneal biomechanics of all participants were measured using Corvis ST machine (Corneal Visualization Scheimpflug Technology; Oculus, Wetzlar, Germany). The machine applanates the cornea using a puff of air, and the corneal deformation process is recorded by a Scheimpflug high-speed camera at a shot speed of 4330 frames/s. During the examination, the participant was seated in a stable position on a chinrest. They were asked to blink and look at a fixed point to expose the cornea. The examiner then adjusted the joystick, and the machine automatically identified the cornea and applied pressure by spraying pulsed air to the front cornea, which produced a series of DCR parameters, waveforms, and videos. DCR parameters were obtained from the built-in Corvis ST software (version 1.6r2187). Only measurements with a quality of "OK" were used for analysis.

2.3. Input data pre-processing

The built-in Corvis ST software was used to obtain the raw data of the deforming corneal during the test. Each data contains two matrices, corresponding to the anterior and posterior surface elevation of the corneas, extracted from 140 2D cross-sectional corneal images during the air-puff induced dynamic deformation. Each matrix was sized at 140 $\times$ 572, i.e., 572 data points across the corneal surface over 140 frames. To address potential challenges arising from missing values in peripheral corneal data and ensure a consistent matrix structure, a 15-degree polynomial fitting procedure was employed for the height data of each frame and 140 locations were sampled from the fitting across the corneal surface, yielding a final matrix size of 140 $\times$ 140 for both anterior and posterior surfaces. To emphasize the corneal deformation features, corneal evelation data at the non-exhalation phase (i.e., the first frame) was used as the baseline corneal shape, which was subsequently subtracted from each successive frame to reach the final data configuration as the input matrix data for the CorNet model (Fig. 1).

2.4. Architecture of CorNet

The architecture of the CorNet comprised four convolutional blocks with a residual structure similar to ResNet [27], albeit with fewer hidden layers (Fig. 2). Benefiting from the residual structure, ResNet allows the neural network to go deeper, be more accurate in classification tasks, have fewer parameters, and solve the gradient problems [29]. Each residual structure includes two parts: basic building blocks and skip connection (Fig. 3). Given an input x, the output features of a residual structure were denoted as ReLU(F(x) + x)). The basic building blocks extracted feature information F(x), and then added it with skip connection x, and used the ReLU activation function to nonlinearly activate F(x) + x. It can also be said that the identity mapping of feature information was achieved by using skip connections. To create a lightweight network and enhance performance, 1 $\times$ 1 convolution was substituted with 3 $\times$ 3 convolution for identity mapping [30]. After the last residual structure, CorNet integrated an adaptive averaging pool layer to reduce the output dimensions to 1 $\times$ 1. It subsequently employed a linear layer to transform the output into a probability distribution, representing the two classes of data (KC vs normal eyes). The selection of the residual structure in CorNet was motivated by its well-structured design for effective feature extraction, alongside its extensive applications and proven performance in various other image recognition tasks [31].

2.5. Training of CorNet

The entire dataset was randomly divided into a training set (70% of the total) for model training and a validation set (30% of the total) for accuracy assessment. To facilitate training, an AdamW optimizer [32] and the cross-entropy loss function were employed. The training set underwent 45 epochs with a learning rate set at 0.001 and a batch size of 16. During training, data augmentation included normalization and 50% horizontal flipping per epoch, alongside a dropout layer (rate 0.25) before the fully connected layer, enhancing accuracy and averting overfitting. Performance assessment in both the training and validation sets relied on overall accuracy as the key metric. An Ubuntu server with Intel(R) Xeon(R) CPU @ 2.20 GHz, and NVIDIA Tesla T4 GPU cards was utilized for training the model.

2.6. Grad-CAM

The Grad-CAM was adopted to visualize the average attention for diagnostic regions. In this visualization, red regions indicate high scores for KC detection, while blue regions indicate lower scores and attention. Grad-CAM is class-discriminative and identifies relevant image regions significant for the final prediction. It uses the gradient of the final convolutional layer to produce an attention map, highlighting the important and class-discriminative areas in the image for the target class detection. Fig. 4 depicts the flowchart of steps to implement Grad-CAM. More details about Grad-CAM can be found in Ref. [28].

2.7. Statistical analysis

Data analyses were performed using the Statistical Package for the Social Sciences software (SPSS version 23; IBM) and the scikit-learn library in Python (version 3.10.6). The means and standard deviations were calculated for the demographic data and Corvis ST metrics. Receiver operating characteristic (ROC) curves for deep learning models of the two categories were calculated to obtain the area under the ROC curves (AUC). Pairwise comparisons of the AUC were accomplished with a nonparametric approach as described by DeLong and associates [33] to compare the performance of diagnostic tests. Statistical significance was defined as $p < 0.05$. Given the true positives (TP), false positives (FP), true negatives (TN) and false negatives (FN) of the two categories, the following metrics were calculated to comprehensively evaluate the classification performances of the deep learning models:

$$Sensitivity = \frac{TP}{TP + FN}$$

$$Specifity = \frac{TN}{FP + TN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

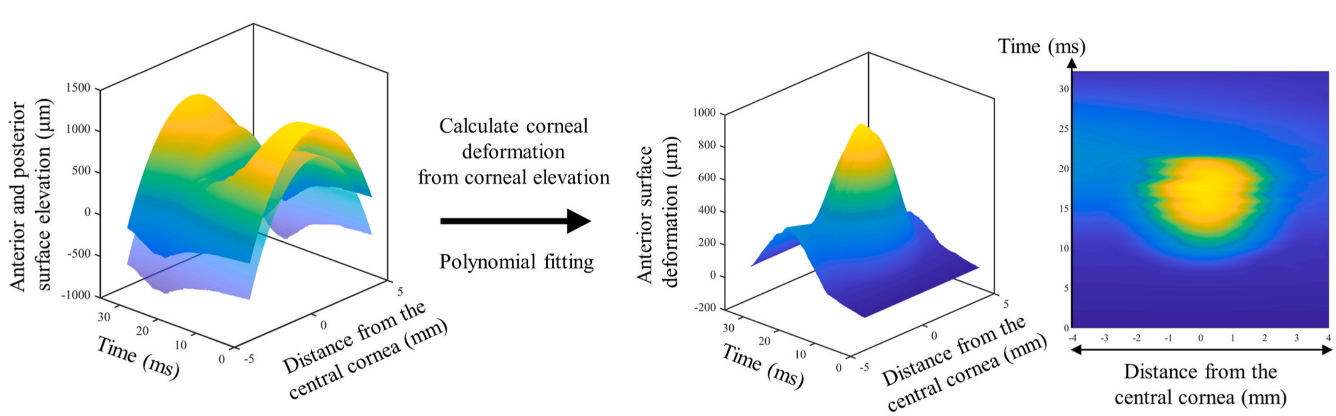

Fig. 1. Schematic plots showing the input data pre-processing.

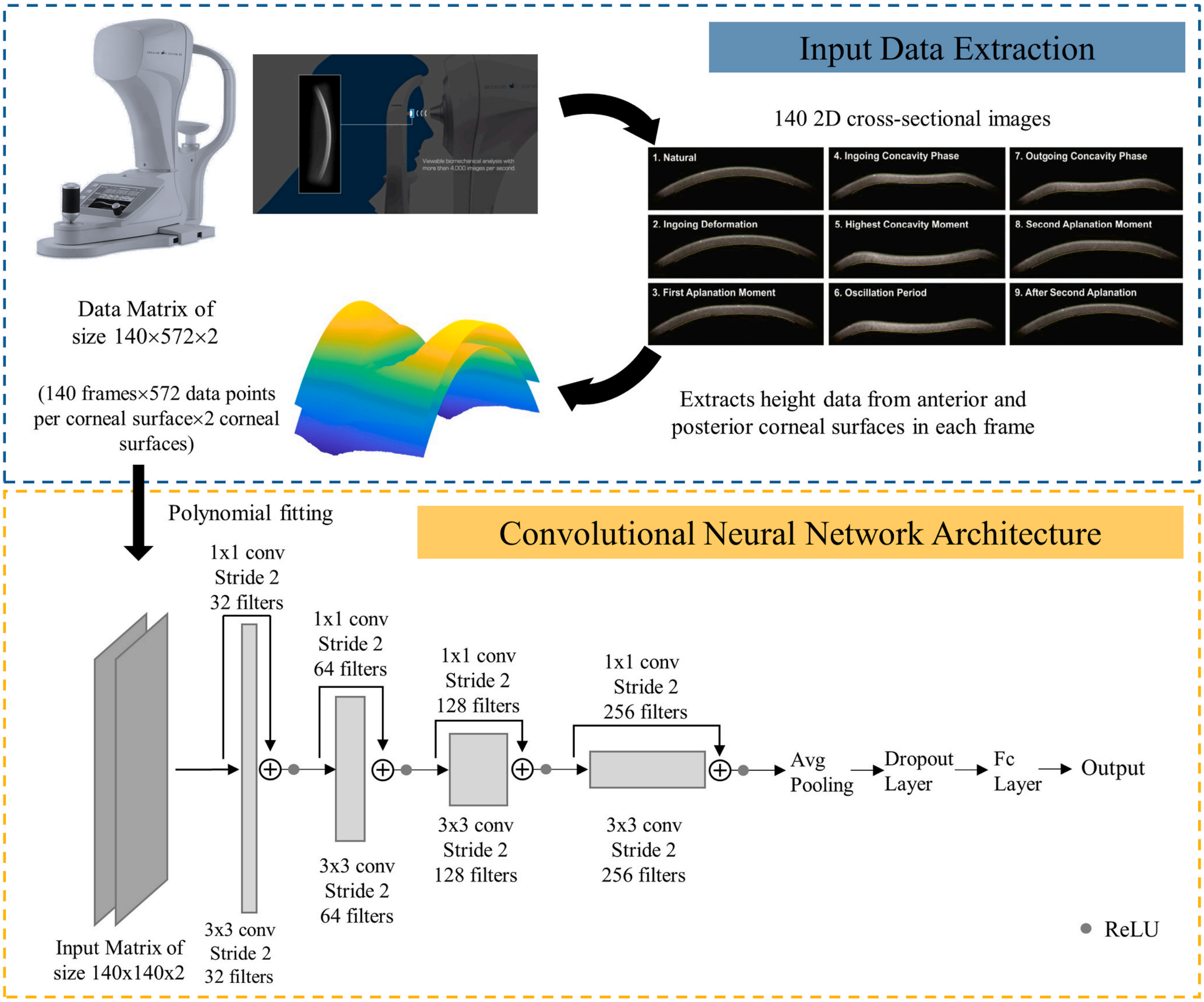

Fig. 2. Method used to create the input matrices and the architecture of the convolutional neural network.

$$F1 = \frac{2 \times Sensitivity \times Precision}{Sensitivity + Precision}$$

3. Result

The demographic information of the study populations is shown in Table 1.

3.1. Corvis ST indices

Table 2 summarizes the diagnostic performance of the top 5 Corvis ST parameters in distinguishing between normal and KC groups (full list given in Supplementary Table 1). These parameters, along with their AUC values, sensitivity, specificity, precision, and accuracy, provided insights into their effectiveness as diagnostic markers. Notably, cCBI exhibited high sensitivity (88.38%), specificity (98.11%), precision (96.58%), and accuracy (94.45%) with an AUC of 0.972, while CBI displayed high sensitivity (89.42%), specificity (94.51%), precision (90.77%), and accuracy (92.60%) with an AUC of 0.964. There was a statistically significant difference between the ROC curves of cCBI and CBI (DeLong $p = 0.0018$).

3.2. Performance of CorNet

The CorNet was evaluated based on both the training set (1252 eyes, 70%) and validation set (534 eyes, 30%). Regarding the overall discriminating power of the validation set, the CorNet model demonstrated a higher overall classification ability (Table 3). In the training set, notable results included cCBI achieving an AUC of 0.975 with high sensitivity (88.98%) and specificity (98.20%), while CorNet demonstrated exceptional performance with an AUC of 0.990, sensitivity (97.30%), and specificity (95.35%). In the validation set, cCBI maintained its strong performance with an AUC of 0.963, high sensitivity (86.43%), and specificity (98.20%), while CorNet excelled with an AUC of 0.971, high sensitivity (92.49%), and specificity (91.54%). These results highlight the effectiveness of cCBI and the CorNet model in distinguishing between normal and KC eyes.

The receiver operating characteristic curves (ROC) are shown in Fig. 5. Within the validation set, notable statistical disparities were evident among the ROC curves. Specifically, a significant distinction was observed between the ROC curves of CorNet and CBI ($p = 0.011$), as well as between cCBI and CBI ($p < 0.001$). However, no significant difference was observed between CorNet and cCBI ($p = 0.295$).

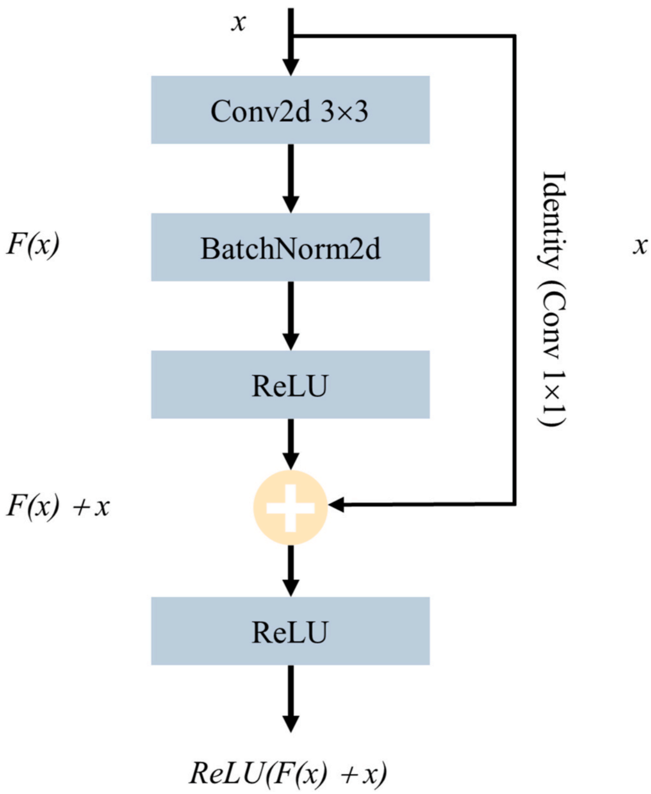

Fig. 3. The residual structure used in this study.

3.3. Grad-CAM visualization

Grad-CAM generated attention maps for all samples correctly identified as KC which illustrate the focal areas of CorNet in distinguishing KC from normal samples. An overall attention map is shown Fig. 6 (A) by averaging all maps. Additionally, Fig. 6(B) displays maps for nine randomly selected samples. In these maps, red areas indicate strong attention of CorNet for KC detection, while blue areas signify low scores and less focus. It is worth noting that the focus areas of CorNet are predominantly observed during the loading phase, which encompasses the transition from 1st Applanation Time (A1T) to Highest Concavity Time (HCT) in the dynamic corneal response, rather than the unloading phase. In addition, from the perspective of spatial distribution, the attention distribution of CorNet is asymmetrical and nonuniform between the nasal and temporal regions.

4. Discussion

In this study, CorNet exhibited robust diagnostic capabilities in discerning between KC and normal eyes, showing superior performance to most existing Corvis ST indices. Notably, this study introduces an end-to-end CNN leveraging raw Corvis ST data. This autonomous feature learning model captures extensive temporal and spatial features, marking a novel approach for KC detection.

The increasing demand for refractive surgery emphasizes the need for more accurate KC detection before ablative procedures to prevent potential complications. The introduction of Corvis ST facilitates KC diagnosis through the assessment of corneal biomechanics using DCR parameters. These metrics exhibit a significant ability to detect the onset of this ectatic disease [34–36], aligning with the results of the current study. Specifically, the investigation identified ARTh as the DCR parameter with the highest diagnostic efficiency, achieving an AUC of 0.946 in the all dataset. However, the cut-off values of DCR parameters for KC detection lack standardization according to different studies, significantly limited their use in clinical practice [17].

To develop standardized values for the parameters obtained, Corvis ST introduced several comprehensive parameters derived from DCR parameters. In particular, previous investigators applied linear regression and random forest algorithms to develop CBI, cCBI, and TBI, achieving significantly enhanced diagnostic performance compared to

Table 1
Basic demographic information.

Parameters [Unit]	Normal	KC
Number [OD/OS]	536/576	342/332
Age [Years]	23.75 ± 5.14	22.24 ± 6.25
Gender[M/F]	285/279	350/124
bIOP [mm Hg]	14.96 ± 2.36	13.27 ± 2.57
CCT [mm]	551.59 ± 32.00	492.43 ± 38.97
SimK [mm]	7.76 ± 0.30	7.29 ± 0.54

Quantitative data are shown as mean ± standard deviation. KC: keratoconus; CCT: central corneal thickness; bIOP: biomechanically corrected intraocular pressure; SimK: simulated keratometry.

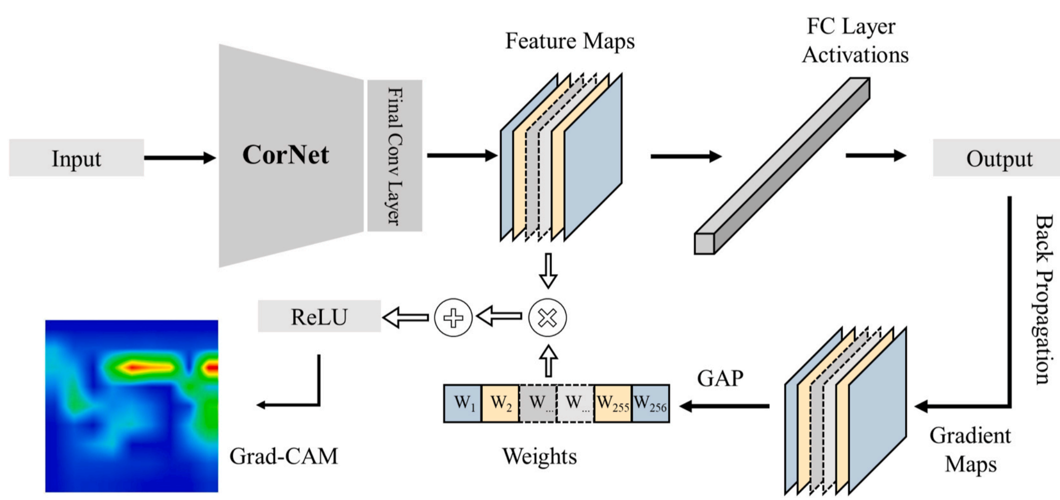

Fig. 4. An overview of gradient-weighted class activation mapping (Grad-CAM). The input data is fed into CorNet to acquire the feature maps from the final convolutional layer along with the model output. Through back propagation, gradient maps are derived from the output. These gradients are used to calculate the GAP for each channel, resulting in values that act as weights for the network. Grad-CAM is generated by applying these weights to the feature maps—multiplying and adding them—then passing the sum through the ReLU. FC Layer: Fully-connected Layer; GAP: global average pooling.

Table 2
Top 5 AUC Corvis ST parameters between normal and KC groups.

Variable	All Dataset					Cut-off Value
	AUC	Sensitivity (%)	Specificity (%)	Precision (%)	Accuracy (%)	
cCBI	0.972	88.38	98.11	96.58	94.45	0.59
CBI	0.964	89.42	94.51	90.77	92.60	0.83
ARTh	0.946	94.33	84.84	91.14	90.76	358.63
IIR	0.922	80.12	89.84	82.70	86.17	9.98
SP-A1	0.921	89.03	81.55	88.87	86.21	84.12

AUC: area under the receiver operating characteristic curve; cCBI: Corvis Biomechanical Index for Chinese populations; CBI: Corvis Biomechanical Index; ARTh: Ambrosio Relational Thickness to the horizontal profile; IIR: integrated inverse radius; SP-A1: stiffness parameter at first applanation.

Table 3
Discriminating performances of the Corvis ST parameters and CorNet model for normal and KC groups.

Dataset	Variable	AUC	Sensitivity(%)	Specificity(%)	Precision(%)	Accuracy(%)	F1(%)
Training Set	cCBI	0.975	88.98	98.20	96.77	94.72	92.72
	CBI	0.971	88.77	96.53	93.95	93.61	91.29
	ARTh	0.949	94.99	85.20	91.36	91.29	93.14
	SP-A1	0.930	89.47	82.84	89.59	86.97	89.53
	IIR	0.924	79.07	91.14	84.42	86.58	81.66
	CorNet	0.990	97.30	95.35	97.18	96.57	97.24
Validation Set	cCBI	0.963	86.43	98.20	96.63	93.80	91.25
	CBI	0.947	88.94	91.59	86.34	90.60	87.62
	ARTh	0.937	90.09	87.50	92.31	89.12	91.19
	IIR	0.916	81.59	88.89	81.59	86.14	81.59
	CBI-LVC	0.907	82.05	88.10	81.22	85.77	81.63
	CorNet	0.971	92.49	91.54	94.77	92.13	93.62

AUC: area under the receiver operating characteristic curve; cCBI: Corvis Biomechanical Index for Chinese populations; CBI: Corvis Biomechanical Index; ARTh: Ambrosio Relational Thickness to the horizontal profile; IIR: integrated inverse radius; SP-A1: stiffness parameter at first applanation.

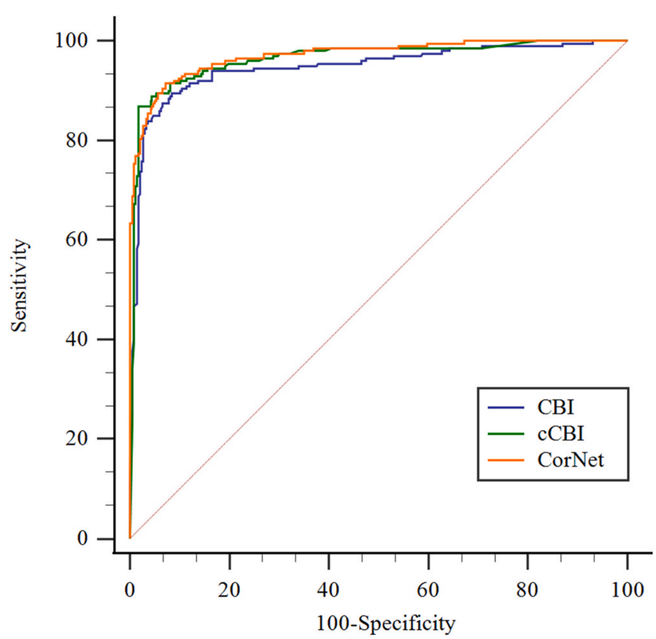

Fig. 5. Receiver operating characteristic curves of cCBI, CBI and CorNet for the validation dataset. cCBI: Corvis Biomechanical Index for Chinese populations; CBI: Corvis Biomechanical Index.

DCR parameters [37–39]. In the current study, both cCBI and CBI exhibit high diagnostic performance with AUC values of 0.972 and 0.964, respectively, ranking among the top parameters for KC detection in existing Corvis ST parameters. The ROC curve analysis showed a significant difference between CorNet and CBI in the validation set but not between CorNet and cCBI. These results emphasize the efficacy of CorNet, showing performance comparable to cCBI and outperforming other Corvis ST parameters, including CBI. Moreover, there was a statistically significant difference between the ROC curve of cCBI and CBI in validation set (Fig. 5, $p < 0.001$). This suggests the improved performance of cCBI over CBI in diagnosing KC in the Chinese population as previously indicated by Vinciguerra et al. [14], signalling the importance for population-based optimization for disagnostic tools.

The progress in ML theory has created opportunities for intelligent KC diagnosis. Numerous studies have harnessed diverse ML algorithms in conjunction with Corvis ST, leading to enhanced KC discrimination. To name a few, the Decision Trees algorithm achieved specificity, sensitivity, and accuracy rates of 98%, 85%, and 92%, respectively, based on parameters extracted from eyeball reactions and corneal deformations during air puff responses [22]. Discriminant analysis, using DA values compensated by IOP, achieved an AUC of 0.954, with a sensitivity of 88.2% and a specificity of 97.4% [23]. Neural Networks demonstrated promising results, attaining an accuracy of 98.7%, sensitivity of 97.4%, and specificity of 100.0% using biomechanical parameters calculated from dynamic corneal deformation videos [24]. And further, a recent study combining Pentacam and Corvis ST data achieved a diagnostic accuracy of 95.5% [40]. Employing random forest models with DCR parameters and corneal thickness-related data from Corvis ST was reported to produce an accuracy of 93% for predicting different KC severity subgroups [41].

Although excellent diagnosis abilities have been demonstrated in these clinical studies mentioned above, these metrics have potential limitations. Most of these studies do not rely on raw data but rather on specific indices extracted from raw data, a process that may lead to potential information loss during feature extraction, Fig. 7.

Feature extraction plays a crucial role in determining classification accuracy by shaping the information content of these features. According to feature extraction methods, handcrafted or specifically engineered features and those derived from deep learning methods are distinguished from each other. Handcrafted features may reduce computational cost, but deep features eliminate the need for task-specific feature engineering, enabling raw input data to encompass comprehensive features [42–44]. CNN, a deep learning algorithm,

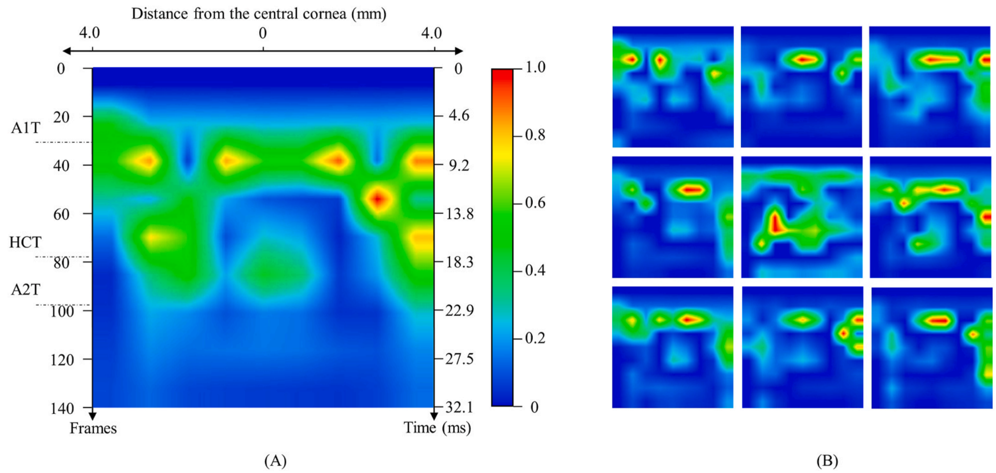

Fig. 6. Visualization of attention maps generated by Grad-CAM. A1T: 1st Applanation Time; A2T: 2nd Applanation Time; HCT: Highest Concavity Time; A1T, A2T, and HCT represent the mean values calculated from the parameters of all KC-predicted samples.

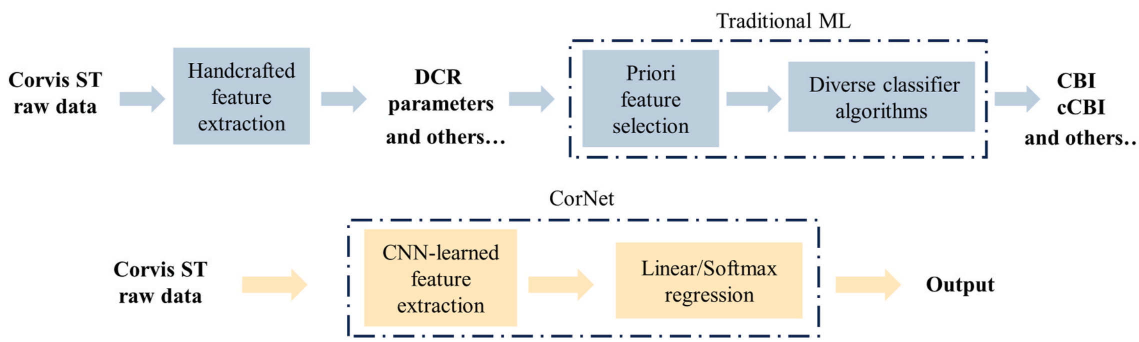

Fig. 7. Work flow chart of traditional ML and CorNet.

combines feature extraction and classification modules into an integrated system (see Fig. 7), allowing CorNet directly processes the 160, 160 raw data points from Corvis ST, autonomously extracting features, thereby avoiding potential information loss linked with handcrafted extraction.

To the authors' knowledge, few studies have utilized a CNN approach with raw Corvis ST data for KC detection. In a parallel study, Abdelmotaal et al. employed a CNN to directly identify KC from pseudoimages extracted from dynamic corneal deformation videos [26]. Their investigation focused solely on contour data from the anterior surface of the cornea, incorporating only the initial 75 frames of the videos. In contrast, the current study incorporated both anterior and posterior corneal surface contour data, utilizing the entirety of the dynamic corneal deformation videos, thereby potentially preventing the loss of valuable information. The demonstrated effectiveness of CorNet and Abdelmotaal et al.'s model in accurately identifying KC underscores the viability and efficiency of integrating CNNs with raw Corvis ST data for diagnostic purposes. The significant advantages of these approaches lie in their ability to maximally combine both spatial and temporal information of the deforming cornea which better reflects the regionally variable and viscoelastic (time-dependent) nature of corneal material properties.

To reveal the inner workings of ML models, Grad-CAM visualization maps are pivotal, particularly in highlighting the spatial and temporal emphasis within the dynamic corneal response for diagnostic purposes. Spatially, the attention distribution of CorNet displays an asymmetric focus between the nasal and temporal regions. Temporally, this attention is predominantly drawn towards the loading phase over the unloading phase, mirroring the highly elastic nature of cornea [16]. These findings are consistent with prior studies, indicating that in KC, elastic deterioration may be stronger than viscoelastic changes [16], and the condition presents as a focal weakness instead of an overall weakened cornea [45]. By utilizing CNN to automatic feature extraction from high-dimensional data and employing an attention visualization algorithm to highlight focused regions, this method aspires to guide the development of additional parameters for these regions, potentially yielding clinical benefits.

A few limitations of this study should be noted. First, the current study did not include pre-clinical KC cases that are more challenging to distinguish from normal eyes. This is mainly due to data scaricity and future study is warranted with richer datasets. Second, the input data used in this study did not fully addressed the influence of whole eye movement during the air-puff test. Considering and correcting for this factor using image registration techniques may further improve the CorNet model, desirable for distinguishing challenging cases such as the pre-clinical ones mentioned above. Third, the CorNet solely relied on Corvis ST data, integrating data from other corneal imaging modalities, such as corneal topographic maps and optical coherence tomography, may lead to enhanced diagnostics. Finally, the cross-sectional design is a limitation. Prospective studies with longer follow-ups are planned to capture different KC stages for early prognosis.

5. Conclusion

This study introduces CorNet, an end-to-end CNN utilizing raw Corvis ST data for KC detection. This novel approach shows diagnostic efficacy surpassing most of existing Corvis ST parameters. The successful fusion of CNN with raw Corvis ST data, autonomously learning comprehensive temporal and spatial features, validates its potential to advance KC diagnosis in ophthalmology.

CRediT authorship contribution statement

PeiPei Zhang: Visualization, Writing – original draft, Conceptualization, Methodology, Software, Validation. **LanTing Yang:** Data curation, Investigation, Writing – original draft, Writing – review & editing. **YiCheng Mao:** Software, Visualization. **XinYu Zhang:** Data curation, Supervision. **JiaXuan Cheng:** Investigation, Methodology, Visualization. **YuanYuan Miao:** Resources, Software, Visualization. **FangJun Bao:** Data curation, Software, Validation. **ShiHao Chen:** Funding acquisition, Supervision, Writing – review & editing. **QinXiang Zheng:** Project administration, Writing – review & editing. **JunJie Wang:** Conceptualization, Funding acquisition, Project administration, Writing – original draft, Writing – review & editing.

Declaration of competing interest

None Declared.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compbiomed.2024.108286.

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